

Certifying Retrieval-Policy Deployment with Non-Neutral AI Judges:

Decision Weights, Control Variates, and What Judge Accuracy Does Not Tell You

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Abstract

Before a retrieval policy is deployed in a new domain, an auditor must certify, from a limited human audit, that it is not worse than the alternatives by more than a tolerance ε . Large language models (LLMs) can now judge relevance at scale, which raises a practical question: *which* AI judge helps the human audit, by how much, and can the certificate still be trusted when the judge is biased toward one of the policies it is judging? We study this question for menus of cutoff policies over a shared ranked candidate pool, with set-F1 or precision utilities, on eight retrieval collections, five AI judges, and a pre-registered external evaluation. Three results follow. (i) The audit savings obtainable from a judge are governed not by its label accuracy but by the correlation ρ between true and judge-predicted *paired policy differences*: over 54 (collection, policy-pair, judge) settings the realised gain of prediction-powered inference (ratio of squared estimated standard errors) correlates 0.78 with ρ and -0.02 with accuracy, the relation holds inside every collection, and a judge built from the same model that defines a policy over-credits that policy by 0.14 F1 on paired differences. A mechanism map shows that savings appear only when $\rho \gtrsim 0.55$, the unlabeled pool is at least twice the audit, and the decision is hard. (ii) In a sample-split certificate the judge enters only as a control variate with mean zero, so a biased judge—including an adversarial one—does not bias the estimator; the coverage of the certificate rests on the bound applied to the judge-corrected differences, not on the judge. That bound is an asymptotically calibrated t bound: its observed wrong-certificate rate is ≤ 0.008 across 24 development cells and 0/500 to 6/500 in three pre-registered evaluations, whereas the earlier in-sample certificate exceeded its nominal risk (0.146 and 0.180 at $\alpha = 0.10$). We also measure what a finite-sample-exact bound costs: on populations of 159–399 queries at $\varepsilon = 0.01$ an exact without-replacement betting certificate does fire, but on two collections only at budgets close to a census (2–5% of runs by the time 90% of the queries are audited), so that it saves almost no labels there, and on the other two in 73–90% of runs by 75%; its range term, not its variance term, sets this cost. (iii) Defining the estimand precisely yields closed-form per-document *decision weights*. Charging every unique human label once, pilot included, importance sampling by these weights roughly halves the labels needed to certify relative to uniform document sampling (-40% to -56% on four collections); a judge used as a λ -fitted control variate adds a further 3–20% of the total cost (16–35% of the labels spent after the pilot) when its ρ is high and nothing beyond noise when it is not; a faithful active-inference rule matches this within noise; the calibrated-judge rule is the one arm that an adversarial judge can break ($+29$ to $+40\%$). Optimising the allocation of labels across the comparisons of a menu pays little: even an oracle minimax design saves 0–14%; every arm’s certification rate is capped by the 8–16% of runs in which the 20-query pilot picks a candidate that cannot be certified, and below that cap the cost is set by the decision margin. We claim no new estimation theory: the contribution is the exact mapping of policy certification onto existing inference tools, the measured conditions under which AI

judges pay, a cost accounting that charges every human label once, and honest reporting of three pre-registered evaluations in which validity replicated every time while the pre-registered efficiency criteria did not—once because the judge’s ρ and the unlabeled pool were too small, once because the locked budget grid was too small for the collection, and once because it was too large and every method saturated.

1 Introduction

Retrieval systems are increasingly deployed as configurable policies: a fixed ranker plus a rule that decides how many documents to return, which threshold to apply, or which reranker to trust. When such a system moves to a new domain, an auditor with a small human-judgment budget must decide whether a candidate policy can be deployed—formally, whether its utility is within ε of the best alternative on the target distribution—or whether to collect more judgments or abstain. Two developments make this problem timely. First, LLM relevance judges are cheap enough to label every candidate document, so the audit can be assisted by a judge. Second, the policies themselves are often built from the same model families as the judges, so a judge may not be neutral toward the policies it evaluates (Balog et al., 2025).

Prediction-powered inference (PPI) and active statistical inference provide the statistical machinery for combining AI predictions with human labels (Angelopoulos et al., 2023a;b; Zrnic & Candès, 2024). Applied naively, however, they leave three questions open that an auditor must answer before spending a budget: *which* judge to use, *whether* the resulting certificate can be trusted if the judge is wrong or biased, and *where* to spend the human labels. This paper answers all three for the policy-certification problem, and reports what happened when the answers were tested on data that had been locked before analysis.

Contributions.

1. **What a judge is worth is not its accuracy.** For paired policy differences, the PPI effective-sample-size gain is $1/(1 - \rho^2)$ where ρ is the correlation of true and predicted differences (Proposition 1). Across 54 (collection, pair, judge) settings the realised gain, measured as the ratio of squared estimated standard errors, correlates 0.78 with ρ and -0.02 with pair-level judge accuracy; the relation holds within every collection and for judge swaps at fixed policies (Section 3). A mechanism map on real data (Figure 2) shows the gain materialises only when $\rho \gtrsim 0.55$, the unlabeled pool is at least twice the audit, and the decision is hard. A judge built from the model that defines a policy over-credits that policy by 0.14 on paired F1 and its ρ for that policy’s comparisons collapses from 0.57 to 0.25 (Section 3.1).
2. **A certificate whose estimator no judge can bias.** Splitting the audit into a training half (policy parameters, judge adoption) and a validation half (the bound) makes the certificate valid conditionally on every choice made on the training half, and the judge enters only through a control variate whose mean is zero, so no judge—however biased—can bias the estimator; the coverage of the certificate depends only on the level of the bound applied to the corrected differences (Proposition 2). Two things are therefore kept apart throughout: the *unbiasedness* of the judge-corrected estimator, which holds for every judge, and the *coverage* of the confidence bound, which depends on the bound’s own hypothesis. The implementation uses a one-sided t bound and is therefore asymptotically, not finite-sample, calibrated; we show empirically that the earlier in-sample certificate exceeded its nominal risk, that the split certificate stays at ≤ 0.008 observed wrong certificates across 24 cells and at 0/500 under an adversarial judge, and that an *always-use-the-judge* policy can lose efficiency with a poor judge while a pilot-based adoption rule protects it. We then price exactness: a finite-population t certificate for the fixed evaluation set stops 20–27% earlier than the pre-registered one, while an exact without-replacement betting certificate is practical only where the margin or the population is large—at $\varepsilon = 0.01$ it fires by 75% of the population in 73–90% of runs on two collections and in 0–2% on the other two, where it fires almost only at budgets close to a census (Section 4).

3. **Where to spend human judgments.** For utilities with a label-independent normaliser, the paired difference is a linear functional with closed-form per-document decision weights (Proposition 3); the common documents carry weight $|1/k_a - 1/k_b|$, so a band-only audit is exact only when cutoffs coincide. With every unique human label charged once (pilot included, one label serving all comparisons of the menu), importance sampling by these weights roughly halves the labels needed to certify relative to uniform sampling on four collections (−40 to −56%); a λ -fitted judge control variate adds 3–20% of the total cost (16–35% of the post-pilot labels) when the judge’s ρ is high and nothing beyond noise when it is not; a faithful active-inference rule matches this within noise; the calibrated-judge rule is the only arm an adversarial judge breaks (Section 5). An exact minimax allocation of labels across the comparisons of the menu, solved with oracle knowledge, saves only 0–14%, and we separate the two reasons (Section 8). For set-F1 we give a bias-corrected linearisation and measure its remainder (Appendix A).
4. **Three pre-registered external evaluations.** On TREC Deep Learning 2021–2023 (211 queries) the query-level methods, on TREC CAsT 2019 (159 conversational turns) the document-level methods, and on ANTIQUE (200 non-factoid questions) the λ -corrected control variate were evaluated under locks committed before the data were touched. Validity replicated every time (wrong certificates 0/500, 0/500 and $\leq 6/500$; boundary type-I error ≤ 0.013). The pre-registered efficiency criteria were not met in any of the three, each time for a reason we can name—too small a ρ and unlabeled pool, too small a budget grid, a saturated one—and we report all three as is (Section 7). The CAsT evaluation exposed a defect of the coefficient-1 control variate under an adversarial judge; the ANTIQUE evaluation confirmed the λ -corrected fix.

What we do not claim. We do not propose a new estimator or a new sampling optimality result; our estimator is a special case of active inference for a linear functional (Section 5.1). We do not claim that AI judges reduce audit cost unconditionally; the pre-registered evaluation shows they need not. We do not claim a general theory of judge neutrality; the self-preference signature we measure is specific to the identical-model judge.

2 Setup

Let P be the target query distribution, $\mathcal{M}$ a finite menu of policies acting on a shared ranked candidate pool for each query q , and $u_m(q) \in [0, 1]$ the utility of policy m on q given human relevance labels $y_d \in \{0, 1\}$ for the pooled documents d . Two utilities are used: set-F1, $u_m(q) = 2 \text{tp}_m / (k_m + n_G)$ with $k_m = |R_m(q)|$ and $n_G(q) = \sum_d y_d$, and precision at the policy’s cutoff, $u_m(q) = \text{tp}_m / k_m$. Population utilities are $\mu_m = \mathbb{E}_P[u_m(q)]$.

Decision and error. A certificate ACT selects a candidate $\hat{m}$ and asserts $\max_j \mu_j - \mu_{\hat{m}} \leq \varepsilon$. A *wrong certificate* is an ACT whose true regret exceeds ε . All methods use the same risk budget: for looks $t = 1..L$ and ordered pairs $(j, \hat{m})$, each one-sided upper confidence bound $U_{t,j}$ on $\mu_j - \mu_{\hat{m}}$ is at level $1 - \alpha'$ with $\alpha' = \alpha / (L |\mathcal{M}| (|\mathcal{M}| - 1))$, and $\text{ACT}_t \iff \max_{j \neq \hat{m}} U_{t,j} \leq \varepsilon$. Throughout $\alpha = 0.10$; $\varepsilon \in \{0.005, 0.01, 0.02, 0.05\}$.

Audit units and the judge. A *query-level* audit labels a query’s whole pool (needed for set-F1 because of n_G); a *document-level* audit labels a sampled subset of documents. An AI judge is a fixed map f from a (query, document) pair to a relevance probability; we threshold it at 0.5 for judge utilities $\hat{u}_m(q)$ and use the probability where a rule requires it. Nothing below assumes the judge is calibrated, unbiased, or independent of the policies.

Efficiency metric. J_{50} is the number of human-judged (query, document) pairs at which the cumulative ACT rate first reaches 50%, including every human label consumed (pilot/training queries, judge selection, residual models, and the bound itself). Between budgets actually run we interpolate linearly; a cell whose largest budget stays below 50% is reported as *not reached* and never enters a ratio. Wrong-certificate rates are reported next to every J_{50} with Clopper–Pearson 95% intervals. Query-level (set-F1) and document-level (precision) blocks use different utilities and are never compared to each other.

3 What determines the value of a judge

Proposition 1 (PPI gain for paired differences). *Fix policies j, m and let $D = u_j - u_m$, $\hat{D} = \hat{u}_j - \hat{u}_m$ with variances $\sigma^2, \hat{\sigma}^2$ and correlation ρ . With n audited and $N \gg n$ unlabeled queries, the PPI++ estimator $\hat{\mu}(\lambda) = \lambda \hat{D}_N + (D - \lambda \hat{D})_n$ has variance $[\sigma^2 - 2\lambda\rho\sigma\hat{\sigma} + \lambda^2\hat{\sigma}^2]/n + \lambda^2\hat{\sigma}^2/N$, minimised at $\lambda^* = \rho\sigma/\hat{\sigma}$ to $\sigma^2(1-\rho^2)/n$. The effective-sample-size gain over the human-only estimator is therefore $1/(1-\rho^2)$: the judge enters only through ρ .*

The proof is a direct variance computation (Appendix B). Writing $\hat{u}_m = u_m + b_m + e_m$ with a policy-specific bias b_m , a bias common to j and m cancels in $\hat{D}$ while a bias that differs between the two policies—for instance one concentrated on documents that only policy j retrieves—lowers ρ regardless of the judge’s overall accuracy.

Empirical test. We measured ρ and the realised gain—the ratio of the squared estimated standard errors of the human-only and PPI++ estimators, averaged over random audits (cross-fitted λ , $T = 10$ audited queries, 300 random audits)—on 54 (collection, pair, judge) settings: four BEIR development pools under a legacy and a modern retrieval stack, five fully judged pools (TREC DL 2019, 2020; TREC-COVID; Touché 2020; DBpedia-Entity), and two judges (Qwen3-Reranker-0.6B and Qwen3-8B with the UMBRELA prompt). Figure 1 shows the result: $\text{corr}(\text{gain}, \rho) = 0.78$ and $\text{corr}(\text{gain}, \text{pair accuracy}) = -0.02$. The relation is not an artefact of pooling heterogeneous groups: demeaned within the 14 (collection, judge) groups it is 0.66 and positive in all 14; and for the 13 (collection, pair) settings in which switching from the reranker to the LLM judge raised ρ , the gain rose in all 13. The estimated-standard-error ratio is a proxy for the quantity of interest, the ratio of the estimators’ mean squared errors about the population gap; on the 33 settings whose pools are in the released bundle (the four BEIR pools under both stacks and the three judged pools with the reranker judge) we re-measured both from 500 random audits per setting. The Monte-Carlo MSE ratio correlates 0.78 with ρ (the estimated-SE ratio 0.83 on the same 33), the two agree to a median absolute difference of 0.03, and where they differ the estimated-SE ratio understates the gain (TREC-COVID, $\rho \approx 0.75$: MSE ratio 1.5–1.7 against 1.1–1.2). The two-sided 90% intervals of both estimators cover the population gap at 0.90 on average (minimum 0.81 over settings, $T = 10$). The 21 remaining settings (TREC DL 2019/2020 and the two LLM-judged BEIR pools) could not be re-measured because their pools and judge outputs are not in the bundle; for them only the estimated-SE ratio is reported.

When does the gain materialise? Proposition 1 is asymptotic in N . On DBpedia-Entity (399 queries) and TREC DL 2021–23 (211 queries) we manipulated three quantities: the judge’s ρ (by replacing a fraction of its labels with base-rate coin flips), the unlabeled size N_{unlab} , and ε , and measured the ACT ratio of the PPI certificate to the human-only certificate at $T = 90$ (Figure 2). The ratio exceeds 1.2 only when all of $\rho \geq 0.55$, $N_{\text{unlab}} \geq 200 \approx 2T$, and $\varepsilon \leq 0.02$ (a decision the human-only certificate cannot settle easily) hold; otherwise it is 0.9–1.05. Every cell has wrong-certificate rate ≤ 0.01 . On TREC DL 2021–23 the judge reaches $\rho \leq 0.46$ and $N_{\text{unlab}} \leq 100$, so no cell shows a gain—which is exactly what the pre-registered evaluation found (Section 7).

3.1 Judges built from the model that defines a policy

We added to the menu a fourth policy, `rr_thresh`, that returns every pooled document whose Qwen3-Reranker score exceeds a threshold tuned on the audit, so that one policy is defined by a model in the judges’ family. With policies fixed at their population parameters, we compared five judges on the six policy pairs of five fully judged pools (Table 1). Two signatures are specific to the *identical-model* judge: it over-credits its own policy by -0.14 on the paired F1 difference (all other judges -0.04 to $+0.03$), and its error rate inside the documents on which the two policies disagree rises from 0.36 to 0.41. The drop in ρ for pairs involving `rr_thresh` appears for every judge, including the different-family MPNet rule, so it is largely a property of that policy’s paired differences rather than of self-preference. Validity is unaffected: with the four-policy menu the split certificate produced 0/500 wrong certificates for both the identical-model and the same-family judge, and the identical-model judge yielded no efficiency gain (ACT 246 vs 245 human-only) while the 8B judge did (328).

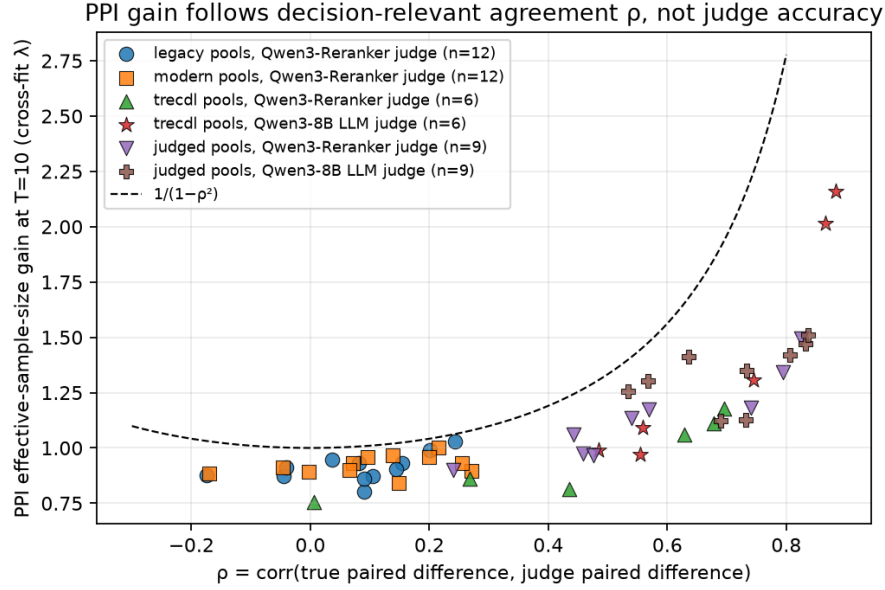


Figure 1: Realised PPI gain (ratio of squared estimated standard errors) at $T = 10$ against ρ , 54 settings, cross-fitted λ . The dashed curve is $1/(1 - \rho^2)$. Colour/marker distinguish pools and judges. Gains track ρ (0.78) and not accuracy (-0.02).

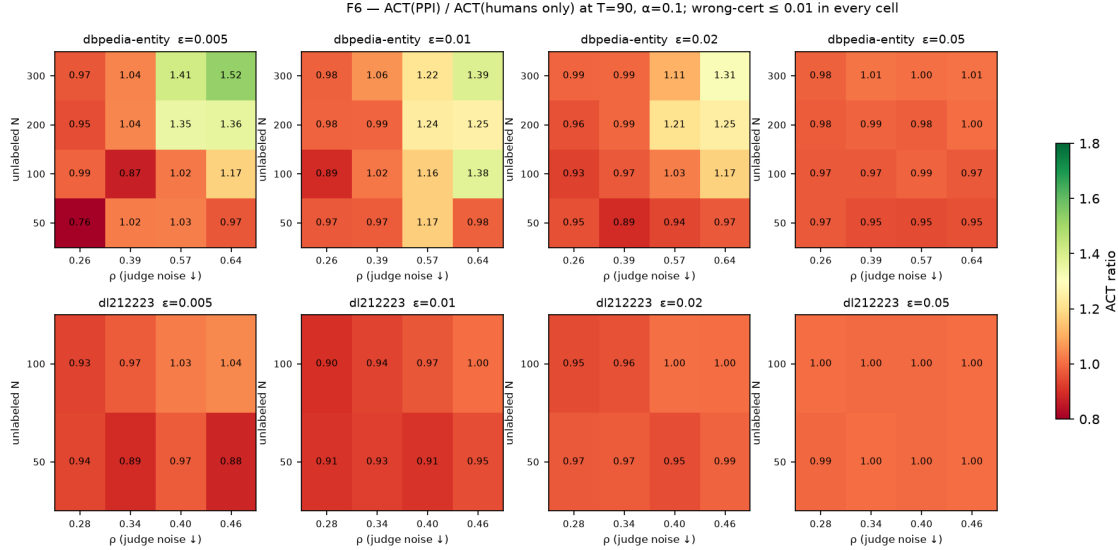


Figure 2: Mechanism map: $\text{ACT}(\text{PPI})/\text{ACT}(\text{humans})$ at $T = 90$ over ρ (judge-label noise), unlabeled pool size and ε ; wrong-certificate rate ≤ 0.01 in every cell. Gains need $\rho \geq 0.55$, $N_{\text{unlab}} \geq 2T$ and a hard decision simultaneously.

4 A certificate whose bias does not depend on the judge

Why the earlier certificate was not. The certificate used in our earlier development (leave-one-out cross-fitted utilities, paired bootstrap, Bonferroni over looks $\times(|\mathcal{M}| - 1)$ comparisons) chose the candidate on the same data used for the bound. With exact t bounds on synthetic data this alone raises the per-look miss rate from the 0.020 budget to 0.049; correcting over all $|\mathcal{M}|(|\mathcal{M}| - 1)$ ordered pairs restores 0.014.

Table 1: Self-preference test at fixed policies: mean over 15 policy pairs per judge, split by whether the pair involves the reranker-defined policy (“without / with”). “Bias” is the mean of predicted minus true paired difference.

Judge	ρ	in-band / out-of-band error	bias	PPI gain
Qwen3-Reranker (identical model)	0.57 / 0.25	0.36 / 0.32 $\rightarrow$ 0.41 / 0.31	−0.03 / − 0.14	1.07 / 0.95
Qwen3-8B (same family)	0.69 / 0.48	0.28 / 0.24 $\rightarrow$ 0.29 / 0.23	0.00 / −0.04	1.19 / 1.04
Mistral-7B (other family)	0.38 / 0.30	0.34 / 0.34 $\rightarrow$ 0.37 / 0.33	+0.02 / −0.04	1.10 / 1.06
MPNet rank rule (other family)	0.39 / 0.21	0.39 / 0.35 $\rightarrow$ 0.40 / 0.35	−0.05 / +0.03	1.03 / 0.94

On real data at 500 repeats the earlier certificate’s wrong-certificate rate was 0.146 [0.116, 0.180] on one development pool and 0.180 [0.147, 0.217] on another, both above $\alpha = 0.10$ with the whole interval.

Split certificate. At look t the audited queries are split into a training half A_t and a validation half B_t . Policy parameters and the decision whether to adopt the judge are made on A_t ; the candidate $\hat{m}_t$ is, in every run reported here, the policy with the largest lower confidence bound on the *validation* half B_t (the implementation can also choose it on A_t ; Proposition 2 covers both and the choice does not affect validity); the bound is computed on B_t only, as a one-sided t bound on the paired differences (humans only) or as a PPI++ bound with λ cross-fitted inside B_t and the unlabeled mean taken outside B_t (with the judge).

Proposition 2 (Simultaneous validity, judge-independent bias). *Fix the menu $\mathcal{M}$ before the audit. Suppose that, for every look t and every ordered pair (j, m) , $U_{t,j,m}$ computed on B_t is a level- $(1 - \alpha')$ upper confidence bound for $\mu_j - \mu_m$ with $\alpha' = \alpha / (T|\mathcal{M}|(|\mathcal{M}| - 1))$. Then on the event $\mathcal{E} = \bigcap_{t,j \neq m} \{\mu_j - \mu_m \leq U_{t,j,m}\}$, which has probability at least $1 - \alpha$, every data-dependent candidate $\hat{m}_t$ —whether chosen on A_t , on B_t , or on both—satisfies $\mu_j - \mu_{\hat{m}_t} \leq U_{t,j,\hat{m}_t}$ for all j ; hence $\Pr(\exists t : \text{ACT}_t \wedge \mu_{\text{best}} - \mu_{\hat{m}_t} > \varepsilon) \leq \alpha$. The statement holds for every fixed judge f , because the judge enters $U_{t,j,m}$ only through a control variate whose expectation is zero; it does not depend on the adoption rule or on whether the ρ -based adoption score covers its target.*

The proof (Appendix B) is a union bound over all ordered pairs and looks, together with the fact that a cross-fitted λ is independent of the fold it is applied to. Two remarks. First, the earlier requirement that the candidate be chosen on the training half is a sufficient but unnecessary condition: what matters is *simultaneous* coverage of the fixed family of $|\mathcal{M}|(|\mathcal{M}| - 1)$ contrasts, not independence of the selector from B_t ; the family itself, however, must not be enlarged after looking at B_t . Second, the hypothesis that each $U_{t,j,m}$ is a level- $(1 - \alpha')$ bound holds only asymptotically for the one-sided t bound used in the pre-registered runs; in finite samples it can fail badly (a paired difference that equals 1 with probability 0.02 and 0 otherwise yields a t bound of 0 whenever no success is observed, and at $n = 30$ this happens on 55% of samples). Empirical-Bernstein and betting bounds (Waudby-Smith & Ramdas, 2024) satisfy the hypothesis exactly for bounded differences; we measured their cost and found them impractical at $\varepsilon = 0.01$ on populations of a few hundred queries (below). The certificate of this paper is therefore an *asymptotically calibrated* one, and the wrong-certificate rates we report are observed Monte-Carlo error rates, not guarantees. The adoption score never enters the bound; a bad adoption decision changes which valid bound is used, hence only efficiency.

Empirical validity. Across 24 (collection, judge, utility) development cells at 500 repeats, the split certificate’s wrong-certificate rate is $\leq 4/500 = 0.008$ (upper 95% limit ≤ 0.020). With an *adversarial* judge (the reranker’s probabilities inverted, pair accuracy 0.34) it is 2/500 and the PPI certificate degenerates to the human-only one (156 vs 158 cumulative ACT at $T = 190$) because $\lambda \rightarrow 0$. Figure 3 shows the cumulative ACT curves by judge on DBpedia-Entity.

What exactness costs. To see whether a finite-sample-exact bound could replace the t bound, we fixed the estimand to the mean over the N queries of the evaluation set (a guarantee about *this* fixed set, not about a future query distribution), so that the audited queries are a sample without replacement and the training half is known exactly: $\mu_N = (\sum_A D + (N - |A|) \mu_{\text{rest}}) / N$ with μ_{rest} bounded from the validation half. Three bounds were compared at $\varepsilon = 0.01$ on the four target collections ($N = 159\text{--}399$, 300 repeats, validation half allowed to grow to $N - 20$): the pre-registered t bound, the same t bound with the finite-population correction and the known training half, and the exact without-replacement betting bound of Waudby-Smith

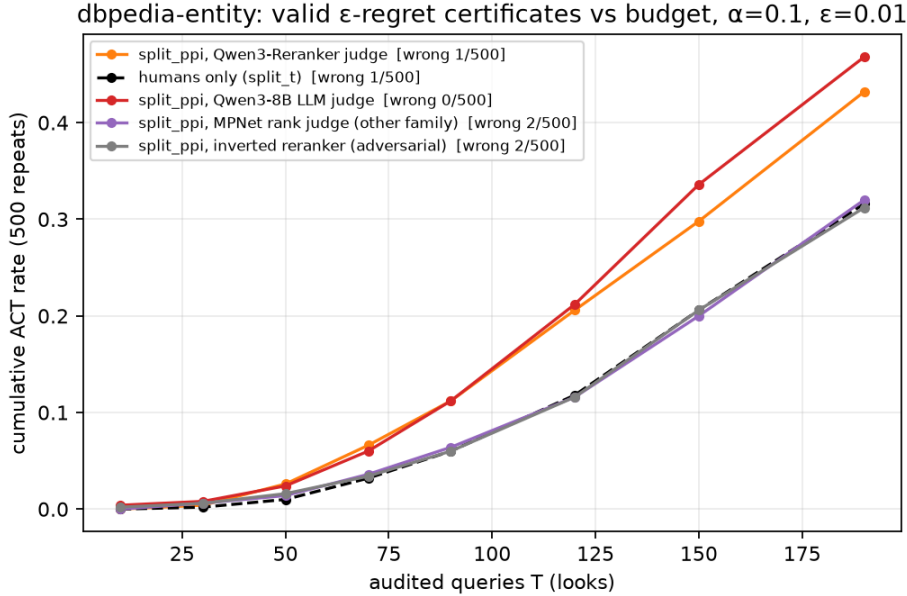


Figure 3: Cumulative ACT rate of the split certificate against the audit budget on DBpedia-Entity (500 repeats, $\varepsilon = 0.01$). Judges change the slope, not the validity: wrong certificates $\leq 2/500$ for every curve, including the inverted (adversarial) judge.

& Ramdas (2020) (Appendix E, Table 7). Every arm was evaluated at every look until it certified or the population was exhausted (an earlier version of this experiment stopped evaluating the exact arms once the t arms had certified, which understated them). The finite-population t certificate is uniformly better than the pre-registered one (mean stopping look 20–27% earlier; ACT 1.00 on all four collections vs 0.65–0.99). The exact betting certificate does fire eventually—in 88–100% of runs—but *when* it fires differs sharply between collections: on DBpedia-Entity ($N = 399$) and DL 21–23 it has certified in 90% and 73% of runs once 75% of the queries are audited, whereas on ANTIQUE and CAsT it has certified in 0–2% by 75% and 3–5% by 90%, and fires almost only at the last look, which audits $N - 1$ queries; there the exact certificate saves essentially no labels over a census. We conjecture that the difference is the certification slack $s = \varepsilon - \text{Regret}_N(\hat{m})$ of the binding comparison together with the population size (the slack enters the without-replacement bound through how far the running mean must stay from ε , the population through how fast the bound shrinks), but residual variance and range also enter, and the attribution needs a fixed-candidate experiment that we have not run. The judge-assisted exact version is worse everywhere because the rectifier’s range is $1 + \lambda$ times wider. The cause is the range term of the bound, not its variance: with paired differences in $[-1, 1]$ and $\alpha' \approx 10^{-3}$, the second-order term $R \log(1/\alpha')/n$ alone exceeds ε until $n \gtrsim 1,400$; removing the union bound over looks (a betting sequence is time-uniform) shortens the bound by only 15%. This says that *this* exact bound is affordable on some collections and close to a census on others, with the slack of the binding comparison as the leading candidate explanation; it is why we keep the t certificate for the main results and report its error rate as observed.

Always using the judge can cost efficiency; a pilot rule protects it. PPI++ with λ estimated on small folds is not guaranteed to be at least as efficient as the human-only bound when the unlabeled pool is small relative to the audit. On TREC DL 2021–23 with the reranker judge (pair accuracy 0.43 there), the always-PPI certificate reached 256/500 cumulative ACT at $T = 90$ against 313/500 for humans only; using the finite- N optimal λ recovered 282, and a variance guard 266, but neither closed the gap. A rule that adopts the judge only when a BCa lower bound on ρ from the training half (at least 30 queries) implies a gain above 1.1 adopted the reranker in 1.3% of runs and stayed at 309. The rule’s value is therefore efficiency protection and the avoidance of judge inference cost; validity does not depend on it. The BCa bound covers

its target at 0.88–0.92 for pilots of 30 queries or more (Appendix F); we call it an *adoption score*, not a guarantee.

5 Where to spend human judgments

Proposition 3 (Decision weights of documents). *For prefix policies a, b with cutoffs $k_a \leq k_b$ on the same ranked pool and a utility $u_m = \sum_{d \in R_m} y_d / Z_m$ whose normaliser does not depend on the labels, $u_a - u_b = \sum_d w(d) y_d$ with $w(d) = 1/Z_a - 1/Z_b$ on common documents, $w(d) = -1/Z_b$ on the band $R_b \setminus R_a$, and 0 elsewhere. Exact band sufficiency holds iff $Z_a = Z_b$; otherwise the common documents carry weight $|1/Z_a - 1/Z_b|$, small when the cutoffs are close and never zero. For set-F1 the normaliser depends on the labels through n_G and the relation holds only after linearisation (Appendix A).*

An earlier draft claimed band sufficiency for precision; a reviewer’s counterexample ($R_A = \{a\}$, $R_B = \{a, b\}$, b irrelevant: the difference is 0.5 or 0 depending on a) showed this is false, and a band-only audit built on that claim yielded no gain at equal budget. Proposition 3 is the corrected statement.

Estimators. Let π_d be the inclusion probability of a human label on document d and $\hat{j}_d$ the judge’s label. The Horvitz–Thompson estimator $\hat{D}_{HT} = \sum_{d \text{ sampled}} w(d) y_d / \pi_d$ and the control-variate estimator $\hat{D}_{CV} = \sum_d w(d) \hat{j}_d + \sum_{d \text{ sampled}} w(d) (y_d - \hat{j}_d) / \pi_d$ are both unbiased for $D(q)$ for any judge; their variances are $\sum_d w(d)^2 y_d^2 (1 - \pi_d) / \pi_d$ and $\sum_d w(d)^2 (y_d - \hat{j}_d)^2 (1 - \pi_d) / \pi_d$. The judge helps exactly on the documents where it is right, weighted by decision weight, and never affects unbiasedness. Query-level certificates (Proposition 2) then apply with $\hat{D}$ in place of a fully audited D .

5.1 Relation to active statistical inference

$\hat{D}_{CV}$ is the active (prediction-powered) estimator of Zrnic & Candès (2024) written for the linear functional $\sum_d w(d) y_d$. Under a budget $\sum_d \pi_d = b$ its variance-minimising allocation is $\pi_d^* \propto |w(d)| \sqrt{\mathbb{E}[(y_d - \hat{j}_d)^2]}$. Our rule $\pi_d \propto |w(d)|$ is the special case of a constant residual model; the “calibrated-judge” rule uses $\hat{j}_d(1 - \hat{j}_d)$; the “residual-estimated” rule fits $\mathbb{E}[(y - \hat{j})^2]$ on the pilot; Li et al. (2025) mix an uncertainty-based rule with a safe distribution, which here is $\pi_d \propto |w(d)|$ itself. What is specific to this paper is not the estimator but the identification of the functional (the decision weights) and the certificate that consumes it.

5.2 Comparison at equal human budget

All arms below share a 20-query fully labelled pilot that fixes policy parameters, the candidate, and any residual model; the same expected number of human-labelled documents per round; and the same certificate. *Cost accounting.* Every human label is charged exactly once through a label ledger keyed by (query, document): the pilot is charged for every document of its 20 queries (not only up to the largest cutoff), one sampled document serves all three comparisons of the menu at once, and nothing is averaged per comparison. This accounting supersedes the per-comparison accounting of our pre-registered runs, which is retained in Appendix C as the historical record; the two differ mainly in how the pilot is charged, which is why the pilot appears as a separate row. *Estimand.* The certificate and the wrong-certificate score target the same quantity, the mean paired difference over all N queries of the collection: the pilot’s contribution is known exactly, and the remaining queries are a without-replacement sample whose per-query document-level estimates carry a two-stage variance, $(1 - f) s_b^2 / n + f \hat{v} / n$ with f the fraction of the remaining queries audited and $\hat{v}$ the Horvitz–Thompson variance of each query’s estimate; when every query is audited ($f = 1$) only document-sampling noise remains. An earlier version of these runs bounded the mean over the non-pilot queries as if they were an i.i.d. sample while scoring the certificate against the population mean; aligning the two lowers every cost, most on CAsT, and is the version reported here. Evaluated on the same draws, the two ingredients contribute unequally: for decision-weight sampling on CAsT the i.i.d. bound needs 3,907 labels, adding the exactly known pilot 2,600 (−33%), and the two-stage variance with the finite-population factor 2,127 (a further −18%); on ANTIQUE the same steps give $881 \rightarrow 851 \rightarrow 836$ (−3%, −2%). The pilot’s share of the population (12.6% on CAsT) and the fraction of the remaining queries audited (all of them at the larger budgets, so that $f = 1$) decide how much each step is worth. Table 2 reports J_{50} for precision at

Table 2: J_{50} under the revised accounting: *unique* human-labelled (query, document) pairs, pilot included, to reach 50% ACT at equal error control (precision at cutoff, $\varepsilon = 0.02$, $\alpha = 0.10$, 300 draws). Observed wrong-certificate rate ≤ 0.003 in every cell (a Monte-Carlo observation, not a guarantee); no cell is censored. Judges: Qwen3-8B (Q), Qwen3-Reranker-0.6B (R), inverted reranker (adversarial, I). Human-only arms do not depend on the judge. Monte-Carlo standard error of a J_{50} cell is about 6–9% (300 draws; slope of the ACT curve at 50%).

Method	ANTIQUE			CAsT 2019			DBpedia-Entity		DL 21–23
	Q	R	I	Q	R	I	Q	R	
Uniform document sampling (humans)	1,405	1,405	1,405	4,778	4,778	4,778	5,563	5,563	4,092
Decision-weight importance sampling (humans)	836	836	836	2,127	2,127	2,127	2,543	2,543	1,783
Stratified by pilot variance (humans)	823	823	823	2,076	2,076	2,076	2,482	2,482	1,785
Active inf., calibrated-judge rule + CV	798	822	1,171	1,986	2,419	2,736	2,520	2,594	2,010
Active inf., residual-estimated rule + CV	753	786	868	1,811	1,970	2,145	2,073	2,273	1,807
Active inf., robust mixture + CV	748	776	886	1,820	2,013	2,182	2,011	2,192	1,822
Decision-weight sampling + judge CV ($\lambda = 1$)	762	794	879	1,788	2,004	2,120	2,084	2,278	1,789
Decision-weight sampling + judge CV (λ fitted)	770	772	822	1,720	1,838	2,024	2,046	2,049	1,722
Pilot labels (common to every arm)	631	631	631	975	975	975	1,029	1,029	1,387

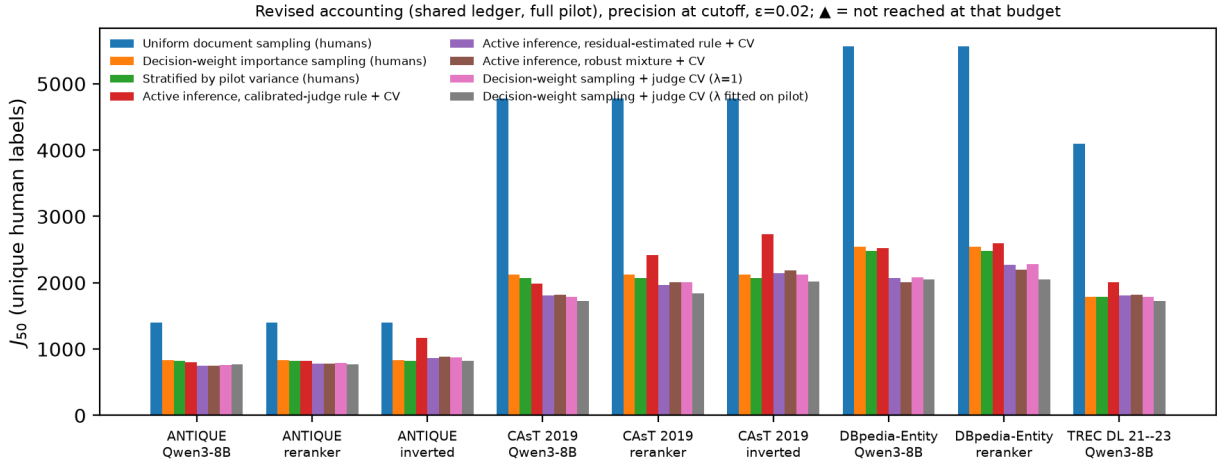


Figure 4: J_{50} by method under the revised accounting (precision at cutoff, $\varepsilon = 0.02$). The bulk of the saving comes from decision-weight importance sampling; the judge adds a judge- and collection-dependent increment as a control variate; the calibrated-judge active rule is the worst judge-assisted arm and the only one that is badly hurt by an adversarial judge.

cutoff at $\varepsilon = 0.02$ on the four fully judged target collections (the two pre-registered document-level targets included), and Figure 4 shows it graphically. Budget grids were extended until every cell was reached without censoring at $\varepsilon = 0.02$; at $\varepsilon = 0.01$ (Table 5, Appendix C) every arm except uniform sampling on CAsT and DBpedia-Entity reaches 50% ACT, with the same ordering.

Four readings follow. First, **the large saving needs no judge**: relative to uniform document sampling, decision-weight importance sampling lowers the per-query estimator variance 2.8–5.0 $\times$ (a ratio the accounting does not touch) and roughly halves J_{50} even with the pilot charged in full—40% on ANTIQUE, –56% on CAsT, –54% on DBpedia-Entity, and –56% on TREC DL 21–23. Stratifying by a pilot-estimated variance adds nothing beyond $|w|$ (0 to –2%, within the Monte-Carlo error of 6–9%). Second, **the judge’s increment is real but modest and conditional**: with the λ -fitted control variate on the same weighted sample, Qwen3-8B lowers J_{50} by 8% (ANTIQUE), 19% (CAsT), 20% (DBpedia-Entity) and 3% (DL 21–23, where its $\rho \approx 0.46$); the reranker by 8–19%; the adversarial judge by 2–5%, i.e. nothing beyond noise. Only the cells at or above 19% exceed twice the Monte-Carlo error. The increment is diluted by the pilot, which is a fixed cost common to every arm: it is 76% and 78% of the weighted arm’s J_{50} on ANTIQUE and DL 21–23 (small populations, large candidate pools) and 46% and 41% on CAsT and DBpedia-Entity;

in units of the labels actually spent after the pilot the Qwen3-8B increment is 32–35% on the first three collections and 16% on DL 21–23. Under the per-comparison accounting of Appendix C the same increments read 14–21%. Third, **the judge can cost labels, and the fix is cheap**: with the adversarial judge the coefficient-1 control variate needs 5% more labels than weighted sampling on ANTIQUE and 0% on CAsT; fitting $\lambda \in [0, 1]$ on the pilot turns this into -2% and -5% (noise) and is never worse than $\lambda = 1$ with the good judges beyond noise. Fourth, **no judge-assisted arm dominates**: the residual-estimated and robust active-inference rules and the two weighted control variates lie within 3–11% of one another in every column (the largest spreads, 10–11%, are with the reranker on CAsT and DBpedia-Entity), whereas the calibrated-judge rule is the only arm that fails badly—+40% over weighted sampling with the adversarial judge on ANTIQUE, +29% on CAsT, +13–14% with the reranker on CAsT and with Qwen3-8B on DL 21–23—because it withholds labels from documents on which the judge is confidently wrong, the failure mode anticipated by Li et al. (2025); with a good judge on an easy collection it is 4–7% cheaper than weighted sampling. We also asked whether allocating labels across the comparisons of a four-policy menu by an exact minimax design helps: an oracle that knows the true gaps and residual variances saves 0–14% of J_{50} over static shared sampling, so we report it as a bounded result (Section 8, Appendix D).

Validity stress tests. Because a wrong-certificate rate of zero is uninformative when the candidate’s true regret is below ε , we ran two stress designs for every document-level arm: forcing the candidate to the true worst policy (type-I error ≤ 0.007 ; an easy test since the gap is then far above ε) and a *boundary* design that sets the candidate to the runner-up and ε to $0.9\times$ its true regret, so that every ACT is a type-I error (realised ε 0.015–0.034): type-I error ≤ 0.03 for every arm on both collections, below α by the Bonferroni margin.

6 Experimental protocol

Pools and policies. Policies are cutoff rules over a shared pool: an expected-F1 gate with a target-calibrated scalar, a global score threshold, a learned truncation, and (for the self-preference test) a reranker-score threshold. Pools are built by the legacy protocol of the base study (BM25 plus three sentence-transformer retrievers, top-30 each, union) and a modern variant with Qwen3-Embedding-0.6B in place of the weakest retriever; the relevance model is a calibrated logistic regression on four pool features trained on four BEIR development collections (nfcopus, scifact, arguana, cqadupstack-android) and never on a target. *Fully judged* pools take the NIST/BEIR-judged documents of each query as the corpus, so every pooled pair carries a human label: TREC DL 2019 (43 queries) and 2020 (54), TREC-COVID (50), Touché 2020 (49), DBpedia-Entity (399), and TREC DL 2021–2023 on MS MARCO v2 (211). The BEIR development pools have sparse judgments; results on them are used only for the ρ analysis and are marked as such.

Judges. Qwen3-Reranker-0.6B (probability of “yes”), Qwen3-8B and Mistral-7B-Instruct-v0.3 with the UMBRELA 0–3 prompt (Upadhyay et al., 2024) read from the next-token distribution over the digit tokens, an MPNet cosine rank rule (label-free, different family), and the inverted reranker as an adversarial control. Batched inference with left padding requires explicit position ids; without them one collection’s judgments collapsed, which we detected and re-ran (Appendix F).

Repeats and randomness. 500 repeats for sequential certificates and 300 draws for fixed-budget designs; every method has its own random stream so adding a method never perturbs another’s numbers; all seeds are recorded.

7 Pre-registered external evaluation

Before downloading TREC DL 2021–2023, we committed a lock fixing the target, pooling rule, training data, $\varepsilon = 0.01$, $\alpha = 0.10$, the look grid, the three methods (humans only, always-PPI, rule-based adoption), the judge, 500 repeats, and three criteria: validity (Clopper–Pearson upper limit of the wrong-certificate rate ≤ 0.15 and point estimate $\leq \alpha$ for all three methods), efficiency (always-PPI $\geq 1.2\times$ humans-only in

Table 3: Pre-registered evaluation on TREC DL 2021–2023 (500 repeats, looks ≤ 90 , $\varepsilon = 0.01$). Cumulative ACT out of 500 and wrong certificates.

Judge	Method	ACT by $T=50$ / 70 / 90	wrong / 500	adoption
—	humans only	84 / 198 / 313	0	—
Qwen3-8B	always-PPI	105 / 220 / 318	0	—
Qwen3-8B	rule-based	84 / 203 / 308	0	26%
Qwen3-Reranker	always-PPI	52 / 136 / 256	0	—
Qwen3-Reranker	rule-based	84 / 194 / 309	0	1.3%
inverted	always-PPI	70 / 182 / 304	0	—
inverted	rule-based	84 / 198 / 313	0	0%
earlier in-sample certificate	—	434 / 494 / 498	7	—

cumulative ACT at the largest look; rule-based $\geq$ humans-only), and adversarial rejection (the rule adopts the inverted judge in $\leq 5\%$ of runs). The pool has 211 queries, so looks are capped at $T = 90$.

Validity passed for every method and judge (0/500; upper limit 0.007), and the adversarial-rejection criterion passed (0%). The efficiency criteria did not: always-PPI reached 318 against 313 ($1.02\times$), and the rule-based method 308. The mechanism map explains the outcome: on this collection the judge’s ρ is 0.40–0.52, the unlabeled pool after auditing is 121 queries, and the policy gaps are large enough that humans alone certify 63% by $T = 90$ —outside the region where Figure 2 predicts a gain. We report the result as pre-registered; the document-level methods of Section 5 were developed after this evaluation and were therefore tested under two further locks, described next.

7.1 Second pre-registered evaluation: document-level methods on TREC CAsT 2019

After the document-level methods of Section 5 were developed, a second lock fixed a target that had never been used: TREC CAsT 2019 evaluation turns restricted to the judged MS MARCO v1 passages (159 turns with a grade- ≥ 2 passage, 48.7 pooled documents per turn, resolved utterances as queries), the same pooling rule and training data, the Qwen3-8B judge, a budget grid of $\{30, 45, 60, 90\}$ full-query equivalents, $\varepsilon \in \{0.01, 0.02\}$, 300 draws, and four criteria: (P1) boundary type-I error $\leq \alpha$ for every arm, (P2) $J_{50}(\text{weighted}) \leq 0.6 J_{50}(\text{uniform})$, (P3) the judge control variate never worse than weighted sampling for three judges (8B, reranker, inverted), and (P4) faithful active inference within 10% of the weighted control variate. The judge reached pair accuracy 0.79 and $\rho = 0.62$, the most favourable judge conditions in this paper.

P1 and P3 passed (maximum type-I error 0.013; judge arms within 0.05 of weighted sampling in ACT at the largest budget). P2 and P4 could not be evaluated as written, because *no* arm reached 50% ACT within the locked budget grid (best 0.43 at $\varepsilon = 0.02$): conversational queries make the policy gaps small, and the grid had been sized on easier collections. By the lock’s own rule this counts as not met, and it is a lesson about pre-registration rather than about the methods—criteria must include a quantity that is always defined, such as ACT at the largest budget. Within the locked grid the ordering of Table 2 replicated exactly: uniform sampling 0.09, decision-weight sampling 0.39, stratified 0.37, calibrated-judge active rule 0.22 (with control variate 0.39), residual-estimated active rule 0.43, decision-weight sampling with the 8B control variate 0.42. A post-hoc extension of the grid to 150 full-query equivalents (reported as post hoc; all J_{50} values in this section use the pre-registered per-comparison accounting, Appendix C; the revised figures are the CAsT columns of Table 2) gives J_{50} of 2,750 judged pairs for decision-weight sampling against *not reached* for uniform sampling, 2,632 with the 8B control variate (-4% , the small increment expected at $\rho = 0.62$), and 2,808 for the residual-estimated active rule. The query-level methods, run for continuity, showed the same picture as Figure 2 predicts for a hard decision with $\rho = 0.62$: cumulative ACT at $T = 70$ of 48/500 for humans only against 100/500 with the 8B judge, 0/500 wrong certificates.

The post-hoc extension also exposed a defect: with the inverted judge the coefficient-1 control variate needed 16% more judgments than weighted sampling alone (3,194 vs 2,750). Unbiasedness does not protect efficiency when the judge is adversarial, exactly as PPI without λ would not. We therefore estimate a

coefficient $\lambda \in [0, 1]$ on the pilot (regression of true on judge paired differences) and use $\lambda\hat{j}$ as the control variate, so that $\lambda \rightarrow 0$ recovers Horvitz–Thompson sampling. This change was made after the lock and its evaluation is post hoc: with the inverted judge the coefficient- λ control variate needs 2,720 judgments against 2,948 for weighted sampling in the same run (coefficient-1: 3,362), and on DBpedia-Entity 2,024 against 2,094 (coefficient-1: 2,388); with the 8B judge it matches or slightly improves on the coefficient-1 version (2,611 vs 2,625; 1,586 vs 1,628). It is the first item of a third lock.

7.2 Third pre-registered evaluation: the λ -corrected control variate on ANTIQUE

The third lock targeted ANTIQUE (Hashemi et al., 2020), a non-factoid question-answering test set never used in development (200 questions, 31.6 pooled answers per question, relevant = grade ≥ 3 of 4), with a wider budget grid $\{30, 60, 90, 120, 150\}$, criteria defined on ACT at the largest budget so that they are always evaluable, and the same judges. The Qwen3-8B judge, prompted on a 0–3 scale that does not match ANTIQUE’s 1–4 scale, reached a pair accuracy of only 0.47 at the fixed threshold—yet $\rho = 0.59$.

Validity passed (P1: boundary type-I error ≤ 0.010). The λ -corrected control variate was never worse than weighted sampling for any of the three judges (P3), it was never worse than the coefficient-1 variate with the inverted judge (P4), and the residual-estimated active rule matched it (P5: 0.94 vs 0.94). The always-defined efficiency criterion (P2: $\text{ACT}(\text{weighted}) \geq 1.5 \times \text{ACT}(\text{uniform})$ at the largest budget) failed for the opposite reason to CAsT: the grid was large enough that every arm saturated (0.96 vs 0.92), whereas the co-registered J_{50} read 776 against 1,555 judged pairs (0.50) under the pre-registered accounting. With the judge, J_{50} fell to 576 (−26%; reranker 627; inverted 733), and at the smallest budget the judge raised ACT from 0.28 to 0.50. Under the revised accounting and the population estimand (Table 2) the same collection reads 836 against 1,405 (−40%) for the human-only arms and 770 with the judge (−8%): the pilot, now charged in full, is 631 of every arm’s labels and shrinks the judge’s relative increment. The coefficient-1 defect reappeared at the 60-equivalent budget with the inverted judge (0.867 vs 0.913 for weighted sampling) and the λ correction removed it (0.930). The query-level certificate, run for continuity, gave 146/500 against 111/500 cumulative ACT at $T = 90$ with 5–6/500 wrong certificates. The lesson for pre-registration is symmetric to CAsT: an always-defined criterion must be evaluated at a budget where the human-only arm is in its informative range, which we now fix by rule.

8 Negative and bounded results

Four things did not work, and we think they are worth knowing. (1) A band-only hybrid predictor (human labels on the disagreement band, judge elsewhere) raises ρ from 0.6 to 0.9 for set-F1 but gives no gain at equal budget, because the rectifier still needs fully labelled queries; for precision it does not raise ρ at all, as Proposition 3 implies. (2) Optimising the allocation of document labels across the comparisons of a four-policy menu pays little, although *sharing* them pays a lot. Two gains must be kept apart. The first is reusing every label for all comparisons of the menu: against independent per-comparison sampling with the same total budget and the same ledger, static shared sampling ($\pi_d \propto \sum_j |a_{jd}|$, the arm of Table 2) needs 43% fewer unique labels on ANTIQUE, 59% fewer on DBpedia-Entity and 35% fewer on DL 21–23 to reach 50% ACT, and on CAsT the per-comparison arm does not reach it at all (Appendix D). The second gain is optimising the shared allocation, and it is the small one. We solved the exact minimax design $\min_{\pi} \max_j \text{Var}_j(\pi)/s_j^2$ (variance of each comparison over its slack to ε , one inclusion probability per document shared by all comparisons) with the true gaps and residual variances, i.e. an oracle. Its design objective is 1.6–2.2 $\times$ better than static shared sampling, yet J_{50} falls by only 0–14% at $\varepsilon = 0.02$ (14% on CAsT, 7% on DL 21–23, 2% on ANTIQUE, 0% on DBpedia-Entity; 1–17% at $\varepsilon = 0.01$), and a plug-in version that estimates gaps and variances from the pilot recovers most of it where the oracle’s saving is large (12% of 14% on CAsT), a third where it is small (2.5% of 7% on DL 21–23) and *loses* 13% on DBpedia-Entity, where the estimation error of the slacks costs more than the design gains. Decomposing the runs separates two causes. (a) In 8–16% of draws the 20-query pilot picks a candidate whose true regret exceeds ε , so no correct certificate exists; every arm’s overall ACT is bounded by one minus that fraction, and on ANTIQUE, CAsT and DL 21–23 both arms reach that cap (conditional ACT among feasible draws 0.97–1.00 at the largest budget). On DBpedia-Entity the conditional ACT is still 0.85–0.89 at 150 query-equivalents,

so part of its plateau is not the pilot but the remaining variance of the hardest comparison. (b) Below the cap the oracle’s advantage is confined to the budgets at which the conditional ACT is far from one: on CAsT at $\varepsilon = 0.01$ it raises the conditional ACT from 0.35 to 0.66 at 90 query-equivalents and from 0.62 to 0.94 at 120, on DL 21–23 by 0.10–0.19 at 5–10 query-equivalents, and by less than 0.05 elsewhere; once the conditional ACT approaches one the design no longer matters. A better variance objective is therefore not a proportionally better certificate: the certificate is a threshold event, and the design changes when it is crossed only for the draws whose slack is of the order of the bound’s width (Appendix D). (3) Two attempts to make always-PPI safe by adjusting λ (finite- N optimum, variance guard) reduced but did not remove its loss with a poor judge; the pilot adoption rule did. (4) An earlier version of this paper reported that judge-uncertainty sampling “harms”; a faithful implementation with a pilot-estimated residual model does not, and we retract the general claim—only the calibrated-judge rule harms.

9 Related work

Prediction-powered and active inference. PPI (Angelopoulos et al., 2023a) and PPI++ (Angelopoulos et al., 2023b) correct AI predictions with human labels and tune λ to bound the loss from poor predictions; active statistical inference (Zrnic & Candès, 2024) chooses which points to label while keeping the guarantee; Li et al. (2025) mix uncertainty-based sampling with a safe distribution; Sfyraki & Wang (2026) find constant-probability sampling competitive for sequential mean estimation; Kilian et al. (2025) give anytime-valid PPI. We use this machinery unchanged. What we add is the functional the certification problem requires (closed-form document weights from a policy pair), a measurement of how the calibrated-judge rule fails on decisive documents, and the observation that a judge used only as a control variate cannot bias the certificate, whatever the judge.

AI judges for IR evaluation. Oosterhuis et al. (2024) build PPI and conformal intervals for IR metrics from LLM judgments; Chatzi et al. (2024) rank LLMs from LLM comparisons with human anchoring; Gligorić et al. (2024) use LLM confidence to allocate human annotation; Balog et al. (2025) document that LLM judges favour LLM-based rankers; UMBRELA (Upadhyay et al., 2024) is the source of our prompt; Durmazkeser et al. (2026) select among many LLMs with few labels. These treat the judge as a fixed predictor of a metric or ranking. We add that for deployment decisions the judge’s value is set by ρ of paired differences, not accuracy; that this varies strongly by judge family and collection (Mistral-7B collapses on Touché while matching Qwen3-8B on DBpedia); and the measured self-preference of the identical-model judge.

Sequential testing and best-arm identification. Adaptive learn-then-test (Zecchin & Simeone, 2024), fixed-confidence BAI (Jamieson & Nowak, 2014; Kaufmann et al., 2016), and ε -best-answer identification (Jourdan & Degenne, 2022) give stopping rules; multi-task PPI (Emmenegger et al., 2026) shares a predictor across tasks. We use a plain union bound over looks and ordered pairs and, in return, handle the winner’s curse of data-driven candidate choice explicitly; the label-sharing structure across comparisons (one document label with different weights per comparison) is outside these frameworks, and our bounded exploration of it was negative.

IR evaluation cost. Topic-set size design (Sakai, 2016), few-topic prediction (Guiver et al., 2009), active judgment sampling (Li & Kanoulas, 2017), active comparison of two models (Sawade et al., 2012), and active testing (Kossen et al., 2021) all trade labels for evaluation precision. We inherit Sawade et al.’s principle of labelling where the difference lives, carry it to the document level with a certification target, and account for costs by audit unit and utility.

10 Limitations and conclusion

Our pools consist of judged documents only, so conclusions concern deployment decisions over judged candidate sets; sparse-judgment collections are used only for the ρ analysis. Precision at cutoff is the primary utility for the document-level results; set-F1 is treated by a bias-corrected linearisation whose remainder we

measure (up to 3ε on one collection). The certificate uses Bonferroni over looks rather than an anytime-valid sequence, and its one-sided t bound is calibrated only asymptotically: the wrong-certificate rates we report (≤ 0.008) are observed over 300–500 Monte-Carlo repeats per cell and are not a finite-sample guarantee; the exact without-replacement bound we tried certifies at affordable budgets on two collections and only near a census on the other two (Section 4), and the finite-population variant that does help certifies the fixed evaluation set only. The cost figures of Table 2 charge every unique human label once, including a 20-query pilot that is 41–78% of the weighted arm’s J_{50} depending on the collection (its share of the menu-allocation arms at their J_{50} is 30–50%); the J_{50} values quoted inside the pre-registered evaluations (Section 7) use the earlier per-comparison accounting and are kept as the historical record (Appendix C); the population sizes (159–399 queries) bound how far any budget curve can be followed, and at $\varepsilon = 0.01$ several cells never reach 50% ACT. The λ -corrected control variate has been tested under one lock (ANTIQUÉ) after being introduced post hoc on CAsT; a further pre-registered evaluation on a collection with a hard decision and a high- ρ judge, where the judge increment should be largest, remains to be done. Three judge families (Qwen3, Mistral, MPNet) were used, with only one LLM judge per family. Within these limits, the paper’s message is a single one: define the estimand that certification actually requires, and the existing tools—importance sampling by decision weight, a judge as control variate, a split certificate—deliver most of what is available: the sampling design roughly halves the human labels, the judge adds a further 3–20% of the total (16–35% of the post-pilot labels) where its ρ is high and nothing where it is not, the judge cannot bias the estimator behind the bound, and neither a better allocation across comparisons nor an exact finite-sample bound changes this picture at the scale of today’s evaluation collections, except on the collections where the exact bound happens to be affordable.

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A Set-F1: linearised document-level auditing

For set-F1, $D(y) = 2\text{tp}_a/(k_a + n_G) - 2\text{tp}_b/(k_b + n_G)$ is a ratio of label sums. We linearise at the judge labels: $w_d = \partial D/\partial y_d|_{y=\hat{j}} = 2 \cdot \mathbf{1}[d \in R_a]/(k_a + \hat{n}_G) - 2 \cdot \mathbf{1}[d \in R_b]/(k_b + \hat{n}_G) - 2\hat{\text{tp}}_a/(k_a + \hat{n}_G)^2 + 2\hat{\text{tp}}_b/(k_b + \hat{n}_G)^2$, so every pooled document has a non-zero weight through n_G ; $\hat{D}_{\text{lin}} = D(\hat{j}) + \sum_{d \text{ sampled}} w_d(y_d - \hat{j}_d)/\pi_d$ with $\pi_d \propto |w_d|$. The remainder $D - \hat{D}_{\text{lin}}$ is measured on the fully labelled pilot and its mean is added as a bias correction with its standard error folded into the bound. A humans-only plug-in alternative estimates $\text{tp}_a, \text{tp}_b, n_G$ by Horvitz–Thompson and plugs them into F1.

Measured remainders: DBpedia-Entity +0.002 (mean absolute 0.011), DBpedia with the reranker judge −0.012 (0.018), TREC DL 21–23 −0.029 (0.030), i.e. three times $\varepsilon = 0.01$ on the last collection, where the uncorrected linearised certificate is not trustworthy. At equal document budget (60 full-query equivalents, $\varepsilon = 0.01$, 300 draws) the ACT rates on DBpedia-Entity were: full-query audit 0.23, plug-in HT (humans) 0.78, query-level PPI 0.33, linearised CV 0.55, bias-corrected linearised CV 0.32; on TREC DL 21–23 (30 full-query equivalents): 0.10, 0.72, 0.10, 0.86 (inflated by the remainder), 0.42. The boundary stress test of Section 5 gave type-I error ≤ 0.013 for all set-F1 arms at realised $\varepsilon \approx 0.03$; it does not probe $\varepsilon = 0.01$ on TREC DL 21–23, where only the bias-corrected variant should be used.

B Proofs

Proposition 1. The two sample means are independent; $\text{Var}[(D - \lambda \hat{D})_n] = (\sigma^2 - 2\lambda\rho\sigma\hat{\sigma} + \lambda^2\hat{\sigma}^2)/n$ and $\text{Var}[\lambda\hat{D}_N] = \lambda^2\hat{\sigma}^2/N$. Minimising the quadratic in λ at $N \rightarrow \infty$ gives $\lambda^* = \rho\sigma/\hat{\sigma}$ and the value $\sigma^2(1 - \rho^2)/n$. $\square$

Proposition 2. Given $\mathcal{G}_t$ (the training half and its decisions), the validation queries are an i.i.d. sample from P independent of $\mathcal{G}_t$. For the human-only bound this is the usual t bound. For the PPI bound, within each fold λ is estimated on the other fold, hence independent of the fold it is applied to, and the unlabeled mean is taken outside B_t ; conditionally on λ , $\mathbb{E}[D_j - \lambda\hat{D}_j] = \mu_j - \mu_{\hat{m}_t} - \lambda\mathbb{E}[\hat{D}_j]$ and $\mathbb{E}[\lambda\hat{D}_N] = \lambda\mathbb{E}[\hat{D}_j]$, so the estimator is conditionally unbiased for any f , with the variance of Proposition 1. Let $\mathcal{E}$ be the event that every one of the $L|\mathcal{M}|(|\mathcal{M}| - 1)$ bounds $U_{t,j,m}$ (all looks, all ordered pairs of the menu fixed before the audit) covers its target; by the union bound $\Pr(\mathcal{E}^c) \leq L|\mathcal{M}|(|\mathcal{M}| - 1)\alpha' = \alpha$ whenever each bound has level $1 - \alpha'$ (exactly for the empirical-Bernstein and betting bounds on bounded differences; asymptotically for the t bound). On $\mathcal{E}$, for any candidate $\hat{m}_t \in \mathcal{M}$, however it was chosen, the pair $(j^*, \hat{m}_t)$ with $j^* = \arg \max_j \mu_j$ belongs to the covered family, so $\mu_{j^*} - \mu_{\hat{m}_t} \leq U_{t,j^*,\hat{m}_t}$; if in addition ACT_t holds then $U_{t,j^*,\hat{m}_t} \leq \varepsilon$, so the candidate is not ε -wrong. Hence a wrong certificate at any look implies $\mathcal{E}^c$. Choosing $\hat{m}_t$ on A_t (an option of the implementation; the reported runs choose it on B_t) is therefore not needed for validity; it matters only if the family of contrasts could be enlarged after inspecting B_t , which the fixed menu rules out. $\square$

Proposition 3. Documents in $R_a \cap R_b$ contribute $y_d/Z_a - y_d/Z_b$, documents in $R_b \setminus R_a$ contribute $-y_d/Z_b$, and documents outside $R_a \cup R_b$ contribute nothing. $\square$

C Pre-registered cost table under the original accounting

Table 4 is the document-level cost table as it stood in our pre-registered runs. Its accounting differs from Table 2 in three ways: documents were averaged per comparison rather than counted once per menu, labels shared by the three comparisons were not de-duplicated (and the per-pair arm drew its own sample for each comparison, which a label ledger later showed cost it more than twice the unique labels of the shared arms), and the pilot was charged only up to the largest policy cutoff rather than for its full pools. The ordering of methods is the same under both accountings; the size of the judge increment is not, because the pilot is a fixed cost common to every arm. We keep the table as the historical record and draw no cost conclusion from it.

Table 4: Original per-comparison accounting (precision at cutoff, $\varepsilon = 0.02$, $\alpha = 0.10$, 300 draws; observed wrong-certificate rate ≤ 0.007). Judge: Qwen3-8B unless stated. Superseded by Table 2.

Method	DBpedia-Entity	DBpedia (reranker judge)	TREC DL 21-23
Uniform document sampling (humans)	n.r. ($> 2,570$)	n.r.	3,207
Decision-weight importance sampling (humans)	1,990	1,990	1,273
Stratified by pilot variance (humans)	2,144	2,144	1,388
Active inference, calibrated-judge rule + CV	2,020	2,054	1,562
Active inference, residual-estimated rule + CV	1,610	1,715	1,340
Active inference, robust mixture + CV	1,556	1,628	1,311
Decision-weight sampling + judge control variate	1,569	1,705	1,256

D Allocation across the comparisons of a menu

For a menu with candidate $\hat{m}$ and $J = |\mathcal{M}| - 1$ competitors, a document-level design is one inclusion probability π_d per document, shared by every comparison, under a budget $\sum_d c_d \pi_d \leq B$. The value of sharing itself is measured by a per-comparison arm that draws an independent sample of B/J documents for each comparison (duplicates across comparisons charged once by the ledger): its J_{50} at $\varepsilon = 0.02$ is 1,784 / n.r. ($\text{ACT} \leq 0.06$ at 4,486) / 5,308 / 3,234 unique labels on ANTIQUE / CAsT / DBpedia-Entity / DL

Table 5: Table 2 at $\varepsilon = 0.01$ (same accounting, estimand and grids). “n.r.” = 50% ACT not reached at the largest budget run (ACT 0.42 on CAsT and 0.38 on DBpedia-Entity). Observed wrong-certificate rate ≤ 0.013 .

Method	ANTIQUÉ			CAsT 2019			DBpedia-Entity		DL 21–23
	Q	R	I	Q	R	I	Q	R	Q
Uniform document sampling (humans)	1,576	1,576	1,576	n.r. (>5,683)	n.r.	n.r.	n.r. (>6,879)	n.r.	4,579
Decision-weight importance sampling (humans)	851	851	851	2,775	2,775	2,775	3,581	3,581	1,958
Stratified by pilot variance (humans)	877	877	877	2,737	2,737	2,737	3,870	3,870	1,979
Active inf., calibrated-judge rule + CV	807	884	1,273	2,459	2,906	3,384	3,979	4,362	2,298
Active inf., residual-estimated rule + CV	781	814	925	2,340	2,500	2,726	2,859	3,345	1,896
Active inf., robust mixture + CV	784	812	943	2,324	2,478	2,682	2,945	3,151	1,869
Decision-weight sampling + judge CV ($\lambda = 1$)	794	817	934	2,320	2,657	2,746	2,959	3,456	1,917
Decision-weight sampling + judge CV (λ fitted)	801	803	870	2,190	2,421	2,633	3,237	3,347	1,883
Pilot labels (common to every arm)	632	632	632	975	975	975	1,028	1,028	1,387

21–23 against 1,015 / 3,181 / 2,186 / 2,102 for static shared sampling, i.e. sharing saves 35–59% where both are measurable. The remainder of this appendix asks how much *optimising* the shared allocation adds on top. Comparison j has estimator variance $V_j(\pi) = \sum_d a_{jd}^2 v_d (1 - \pi_d) / \pi_d$ (a_{jd} its decision weight, v_d the residual variance of the control variate) and slack $s_j = \varepsilon - (\mu_j - \mu_{\hat{m}})$; the certificate fires when every V_j is small relative to s_j^2 , so the natural design is $\min_\pi \max_j V_j(\pi) / s_j^2$. Its dual over the simplex has a closed-form water-filling inner solution, $\pi_d = \min\{1, \sqrt{g_d / (\mu c_d)}\}$ with $g_d = v_d \sum_j \theta_j a_{jd}^2 / s_j^2$, and we solve the outer problem by exponentiated gradient. Two versions were run: an *oracle* that knows the true slacks and the population residual-variance function (but not the labels), and a *plug-in* that estimates both from the pilot and the labels accumulated so far. Both were compared with static shared sampling $\pi_d \propto \sum_j |a_{jd}|$ (the arm of Table 2), its variance-aware variant $\pi_d \propto \sum_j |a_{jd}| \sqrt{\hat{v}_d}$, and an adaptive re-weighting of comparisons by their estimated slack, all with the same ledger, pilot, and certificate, on the four target collections at 300 draws.

Table 6: Menu allocation, $\varepsilon = 0.02$ and 0.01 , Qwen3-8B judge, population estimand, 300 draws. “Design ratio” is the realised $\max_j V_j / s_j^2$ of static shared sampling divided by the oracle’s (structural head-room, independent of the bound); “saving” is $1 - J_{50}(\text{arm}) / J_{50}(\text{static})$ in unique labels (Monte-Carlo error 6–9%); “infeasible” is the fraction of draws in which the pilot’s candidate has true regret $> \varepsilon$; “ACT |feasible” is the certification rate among the other draws at the largest budget run; “non-overlap” is the share of the binding comparison’s decision weight on documents that no other comparison weights. Observed wrong-certificate rate ≤ 0.003 .

Collection	ε	design ratio	saving (oracle)	saving (plug-in)	infeasible	ACT feas. (static / oracle)	non-overlap
ANTIQUÉ	0.02	1.6	2%	2%	0.15	0.98 / 1.00	0.59
CAsT 2019	0.02	2.0	14%	12%	0.08	1.00 / 1.00	0.60
DBpedia-Entity	0.02	1.6	0%	−13%	0.08	0.85 / 0.89	0.56
TREC DL 21–23	0.02	2.2	7%	2%	0.13	0.98 / 1.00	0.83
ANTIQUÉ	0.01	1.6	1%	−1%	0.14	1.00 / 1.00	0.58
CAsT 2019	0.01	2.2	17%	13%	0.15	0.97 / 1.00	0.60
DBpedia-Entity	0.01	1.6	1%	−12%	0.16	0.83 / 0.87	0.55
TREC DL 21–23	0.01	2.1	6%	3%	0.13	1.00 / 1.00	0.83

Three facts explain the gap between the design ratio and the saving, and they do not weigh equally on every collection. (i) *The pilot’s candidate choice caps every arm.* With a 20-query pilot, 8–16% of draws select a candidate whose true regret exceeds ε ; no allocation can certify these correctly. On ANTIQUÉ, CAsT and DL 21–23 the conditional ACT among feasible draws reaches 0.97–1.00 for both arms at the largest budget, so the observed plateau is this cap and nothing else. On DBpedia-Entity the conditional ACT is 0.83–0.89 at 150 query-equivalents and still rising, so part of its plateau is residual variance of the hardest comparison, not the pilot. (ii) *The oracle’s advantage lives only in the transition.* It is largest where the conditional ACT is far from its cap and the budget is binding: CAsT at $\varepsilon = 0.01$ (+0.31 and +0.32 in conditional ACT at 90 and 120 query-equivalents, which is the 17% saving), DL 21–23 at 5–10 query-equivalents (+0.10–0.19); elsewhere it is below 0.05. Two facts about CAsT make it the collection where the design matters: its policies are close (median slack $\approx \varepsilon$, 7–15% of feasible draws with slack below $\varepsilon/2$) and its comparisons’ decision

weights overlap little (non-overlap 0.60), so concentrating labels on the binding comparison’s documents is both needed and possible. (iii) *The pilot dilutes the saving*. The pilot is 30–50% of the labels at J_{50} for these arms and is identical across them, so a proportional saving on the sampled labels appears as a smaller saving on the total. Under the earlier version of these runs, which bounded the mean over the non-pilot queries rather than the population mean, CAsT did not certify at all at these budgets; with the estimand aligned it does, and the ranking of the arms is unchanged. We conclude that, at these population sizes and with a pilot of this size, the certification cost is governed first by the pilot’s candidate choice and the decision margin and only then by the allocation of labels across comparisons, with CAsT as the one collection where the allocation is worth its cost.

E Exact and finite-population certificates

The estimand is fixed to the mean over the N queries of the evaluation set; the training half A of the audit is fully labelled and therefore known exactly, and the validation half B is a uniformly random sample without replacement from the remaining $N - |A|$ queries, so that $\mu_N = (\sum_{q \in A} D_q + (N - |A|)\mu_{\text{rest}})/N$ and only μ_{rest} needs a bound. Three bounds on μ_{rest} were compared for the human-only certificate: the pre-registered one-sided t bound on B alone (targeting the superpopulation mean, no use of A); the same t bound with the finite-population correction $\sqrt{1 - |B|/(N - |A|)}$ inside the decomposition above; and the exact without-replacement betting bound of Waudby-Smith & Ramdas (2020), in which the hypothesis “ $\mu_{\text{rest}} = m$ ” is tested by a nonnegative supermartingale whose predictable conditional mean after i draws is $(N'm - S_i)/(N' - i)$, and hypotheses that become impossible given the observed prefix are rejected outright. The judge-assisted version bounds the rectifier $D_q - \lambda \hat{D}_q$ (range $[-1 - \lambda, 1 + \lambda]$, $\lambda \in [0, 1]$ fitted on A) and adds the exactly known population mean of $\lambda \hat{D}_q$. The training half was fixed at 20 queries so that the validation half could grow to $N - 20$; every arm was evaluated at every look until it certified or the last look was reached (the last look audits $N - 1$ queries). Level $\alpha' = \alpha/(L|\mathcal{M}|(|\mathcal{M}| - 1))$ with $L = 14$ looks, $\varepsilon = 0.01$, 300 repeats; wrong certificates were scored against the population mean. T counts all audited queries, the 20-query training half included, so T/N is the share of the population labelled.

Table 7: Split certificates under three bounds, population estimand, $\varepsilon = 0.01$, 300 repeats. t columns: certification rate and mean stopping look $\bar{T}$ (queries audited, pilot included). Betting columns: cumulative certification rate by the time 75% / 90% / 100% of the population has been audited, and $\bar{T}$ for the human-only arm. The last column is the largest look run. Observed wrong-certificate rate ≤ 0.007 (t) and 0 (betting) in every cell.

Collection (N)	t , pre-reg.	t +FPC+known A	t +FPC+judge	bet. WoR (75/90/100%)			bet. WoR+judge	max T/N
ANTIQUA (200)	0.65 ($\bar{T}$ =148)	1.00 ($\bar{T}$ =110)	1.00 ($\bar{T}$ =103)	0.02 / 0.03 / 0.88	($\bar{T}$ =198)	0.00 / 0.01 / 0.71		100%
CAsT 2019 (159)	0.77 ($\bar{T}$ =121)	1.00 ($\bar{T}$ =88)	1.00 ($\bar{T}$ =79)	0.00 / 0.05 / 0.91	($\bar{T}$ =152)	0.00 / 0.01 / 0.73		99%
DBpedia-Entity (399)	0.94 ($\bar{T}$ =142)	1.00 ($\bar{T}$ =113)	1.00 ($\bar{T}$ =96)	0.90 / 0.95 / 1.00	($\bar{T}$ =218)	0.56 / 0.94 / 1.00		100%
TREC DL 21–23 (211)	0.99 ($\bar{T}$ =92)	1.00 ($\bar{T}$ =71)	1.00 ($\bar{T}$ =68)	0.73 / 0.73 / 1.00	($\bar{T}$ =161)	0.47 / 0.47 / 1.00		100%

A synthetic check isolates the cause. For $N = 399$, paired differences with mean -0.02 and standard deviation 0.05 in $[-1, 1]$, and $\alpha' \approx 10^{-3}$, the betting bound at $n = 100$ is $+0.06$ while the t bound is -0.004 ; the betting bound crosses $\varepsilon = 0.01$ only at $n \approx 250$ (63% of the population), and this is insensitive to the resolution of the candidate-mean grid. With range $R = 2$ the second-order term $R \log(1/\alpha')/n$ of any empirical-Bernstein-type bound is 0.14 at $n = 100$, fourteen times ε ; halving the range would bring the crossing to $n \approx 100$, but no such range is known a priori for paired precision differences. Dropping the union bound over looks, which a betting sequence does not need, lowers $\log(1/\alpha')$ from 7.1 to 4.8 and the width by about 15%. The runs of Table 7 show the same thing on real data: on DBpedia-Entity ($N = 399$) and DL 21–23 the betting certificate has fired in 90% and 73% of runs by the time 75% of the population is audited, and on ANTIQUA and CAsT it fires almost only at the last look ($N - 1$ queries audited), where the unobserved remainder is a single query and the bound is nearly the population mean itself; on those two collections exact certification and a census cost nearly the same. Whether the difference is the slack $s = \varepsilon - \text{Regret}_N(\hat{m})$ of the binding comparison, the population size, or the residual variance is not settled by these runs. The finite-population t certificate, by contrast, is a strict improvement on the pre-registered one

at no cost in observed error, and we recommend it whenever the guarantee is meant for the fixed evaluation set; it must not be read as a guarantee about a future query distribution.

F Reproducibility notes

All numbers are regenerated from row-level result files by the scripts in the repository (the revised cost table by `86_table2_v2.py` from the `_v2` result files; the menu-allocation and exact-certificate appendices from `83_menu_allocation.py` and `63_planner_v2.py` with the options recorded in the design log). Each method uses an independent random stream derived from the repeat index and the method name. Batched causal-LM inference with left padding must pass explicit position ids; without them the relevance judgments on one collection (TREC-COVID) collapsed to a constant, which we detected from a 49/50-query anomaly, fixed, and re-ran for every collection (label changes $\leq 0.5\%$ elsewhere; reranker scores unaffected, checked against batch-size-1 scoring). The BCa lower bound on ρ used by the adoption rule covered its target at 0.88–0.92 for pilots of 30 and 50 queries on DBpedia-Entity (0.86 at 20), against 0.75–0.84 for the Fisher- z bound on skewed pairs. The lock document for Section 7 was committed before the target data were accessed; the only pre-lock access was the count of judged queries and pairs.